LEGO Design Enrichment Program - Term 3

8-Week Curriculum Guide

Think. Play. Grow.

Beginner Level

Creative exploration: Fantasy building, problem-solving, and group projects

Week 1: Fantasy Creatures

Component	Description
Learning Goals	• Design mythical creatures • Combine animal features • Add magical elements
Activities	• Dragon with moving wings • Unicorn and phoenix builds • Creature feature design
Superpower	Mythical Imagination - Creating legends

Week 2: Amusement Park

Component	Description
Learning Goals	• Build carnival rides • Create game booths • Design entertainment venues
Activities	• Ferris wheel construction • Roller coaster design • Midway games
Superpower	Entertainment Design - Creating fun

Week 3: Emergency Vehicles

Component	Description
Learning Goals	• Build police cars and fire trucks • Add emergency features • Create action scenes

Component	Description
Activities	• Fire truck with ladder • Police station design • Ambulance and rescue vehicles
Superpower	Heroic Building - Creating helpers

Week 4: Jungle Adventure

Component	Description
Learning Goals	• Build tropical environments • Create jungle animals • Design exploration scenes
Activities	• Rainforest tree building • Monkey and tiger creation • Temple ruins exploration
Superpower	Adventure Creation - Building excitement

Week 5: City Services

Component	Description
Learning Goals	• Build essential city buildings • Create service vehicles • Understand community helpers
Activities	• Post office and library • Garbage truck and street sweeper • Community building
Superpower	Community Understanding - Building society

Week 6: Winter Wonderland

Component	Description
Learning Goals	• Build snowy scenes • Create winter activities • Use white and blue effectively
Activities	• Ski slope construction • Igloo and snow fort • Ice skating rink

Component	Description
Superpower	Seasonal Building - Capturing seasons

Week 7: Music & Performance

Component	Description
Learning Goals	• Build musical instruments • Create performance stages • Design concert venues
Activities	• Guitar and drum builds • Stage with lights • Concert hall creation
Superpower	Performance Space - Building for entertainment

Week 8: Dream Project Day

Component	Description
Learning Goals	• Build anything imagined • Complete ambitious projects • Share creative visions
Activities	• Open-ended dream builds • Support and collaboration • Dream showcase celebration
Superpower	Unlimited Creativity - Building dreams

Intermediate Level

Engineering focus: Structural challenges, advanced mechanisms, and problem-based learning

Week 1: Crane Engineering

Component	Description
Learning Goals	• Build functioning cranes • Create lifting mechanisms • Design stable bases
Activities	• Tower crane construction • Mobile crane building • Lifting competition
Superpower	Lift Engineering - Defeating gravity

Week 2: Renewable Energy

Component	Description
Learning Goals	• Build wind turbines • Create solar panel arrays • Design eco-friendly structures
Activities	• Wind farm construction • Solar power station • Green building design
Superpower	Sustainability - Building for the future

Week 3: Robotics Basics

Component	Description
Learning Goals	• Introduce programmable elements • Build robot chassis • Create simple programs
Activities	• Robot base construction • Motor placement • Simple programming tasks
Superpower	Programming Basics - Controlling machines

Week 4: Medieval Siege Weapons

Component	Description
Learning Goals	• Build catapults and trebuchets • Understand mechanical advantage • Test projectile motion
Activities	• Catapult construction • Trebuchet engineering • Siege weapon competition
Superpower	Physics Application - Using leverage

Week 5: Skyscraper Engineering

Component	Description
Learning Goals	• Build tall stable structures • Understand center of gravity • Design vertical transportation
Activities	• Tall building competition • Elevator system design • Earthquake resistance testing
Superpower	Vertical Building - Reaching heights

Week 6: Transportation Systems

Component	Description
Learning Goals	• Design complete transport networks • Build stations and terminals • Create vehicle fleets
Activities	• Airport or train station • Bus and subway system • Integrated transport planning
Superpower	Systems Thinking - Connecting elements

Week 7: Art Installation

Component	Description
Learning Goals	• Create abstract art pieces • Build for visual impact • Express artistic concepts

Component	Description
Activities	• Abstract sculpture • Installation art pieces • Gallery-quality builds
Superpower	Abstract Expression - Art through bricks

Week 8: Innovation Challenge

Component	Description
Learning Goals	• Solve real-world problems • Pitch creative solutions • Build prototypes
Activities	• Problem identification • Solution brainstorming • Prototype building and presentation
Superpower	Problem-Solving - Creating solutions

Advanced Level

Expert level: Complex systems, artistic mastery, and professional portfolio development

Week 1: Clockwork Mechanisms

Component	Description
Learning Goals	• Build functioning clock movements • Create escapement mechanisms • Master precision timing
Activities	• Clock face construction • Gear timing systems • Pendulum mechanisms
Superpower	Precision Timing - Mechanical accuracy

Week 2: Naval Architecture

Component	Description
Learning Goals	• Design ship hulls • Build maritime vessels • Create naval fleets
Activities	• Aircraft carrier building • Submarine design • Historic ship recreation
Superpower	Maritime Design - Building for water

Week 3: Landscape Architecture

Component	Description
Learning Goals	• Design outdoor spaces • Create terrain features • Build park systems
Activities	• Topographical building • Garden design • Public space planning
Superpower	Environmental Design - Shaping land

Week 4: Steampunk Design

Component	Description
Learning Goals	• Master Victorian-tech aesthetic • Combine gears and pipes • Create anachronistic machines
Activities	• Steampunk vehicles • Mechanical contraptions • Victorian-inspired builds
Superpower	Aesthetic Synthesis - Blending styles

Week 5: Competition Preparation

Component	Description
Learning Goals	• Prepare for competitions • Build under strict rules • Perfect presentation skills
Activities	• Competition rule study • Timed building practice • Presentation rehearsal
Superpower	Competition Excellence - Winning performances

Week 6: Professional Photography

Component	Description
Learning Goals	• Photograph builds effectively • Use lighting and angles • Create portfolio quality images
Activities	• Photography techniques • Lighting setup • Portfolio building
Superpower	Visual Documentation - Capturing work

Week 7: Teaching & Instruction

Component	Description
Learning Goals	• Create building instructions • Teach building techniques • Mentor younger builders

Component	Description
Activities	• Instruction manual creation • Workshop leading • Mentorship sessions
Superpower	Instruction - Teaching others

Week 8: Portfolio Presentation

Component	Description
Learning Goals	• Compile year's work • Present to audience • Demonstrate growth
Activities	• Portfolio curation • Professional presentation • Portfolio showcase event
Superpower	Professional Presentation - Showing expertise